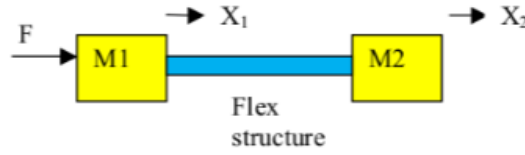


TEK4090 – Introduction to Modern Control

Assignment 2 v1.0

Due: November 17 2025, 23:30

- (40 points) Consider the following model of two masses connected by a flexible structure, with a lateral force on one of the masses.



A dynamical system can be written as,

$$(M_{lump} + M_{rod})\ddot{U} + C_{rod}\dot{U} + K_{rod}U = \begin{bmatrix} 1 \\ 0 \end{bmatrix} F, \quad (1)$$

where $U = [x_1, x_2]^T \in \mathbb{R}^2$ is the displacement of the two masses. Given the density ρ_r of the rod, the elastic modulus E_r of the rod, the cross-sectional area A_r of the rod, the length L_r of the rod, and the masses M_1 and M_2 , the system matrices can be written as,

$$M_{lump} = \begin{bmatrix} M_1 & 0 \\ 0 & M_1 \end{bmatrix}, \quad M_{rod} = \frac{A_r L_r \rho_r}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (2)$$

$$C_{rod} = \alpha K_{rod}, \quad K_{rod} = \frac{A_r E_r}{L_r} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \quad (3)$$

For simplicity, let

$$\frac{A_r L_r \rho_r}{6} = 1; \quad \frac{A_r E_r}{L_r} = 1; \quad M_1 = M_2 = 10; \quad \alpha = 0.1. \quad (4)$$

Finally, let the input be F and the output be x_2 .

- Obtain the state space model $\Sigma := (A, B, C, D)$ of the system. Use Matlab to find the poles and zeros of the system.
- Use Matlab to find the transfer function of the system, and verify the locations of the poles and zeros
- The desired output acceleration of the system is $\ddot{y} = A \sin(\omega t)$. Choose A such that the output reaches 1 after one period $T = 10s$, and set it so that the acceleration reaches zero after T . Finally, obtain and plot the desired velocity and position as a function of time from $t_0 = 0$ to $t_f = T$.

- (d) Choose your favourite feedback system to track the desired reference output (for example, LQR) and explain your design decisions.
 - (e) Simulate and plot the output tracking of the system with this feedback (for example, using LSIM in Matlab). What happens to the tracking performance if T is varied?
 - (f) Find the internal dynamics of the system, then find the reference state trajectory x_{ref} and the inverse input u_{inv} . Plot them, and describe how you found them.
 - (g) Simulate and plot the system with this feedforward input.
 - (h) Add the feedforward and feedback inputs together, and simulate the results in Matlab. Which setup has the best tracking performance (i.e., feedback, feedforward, feedback + feedforward)?
 - (i) Using the feedback + feedforward controller in the previous part, modify the system slightly before simulation (for example, modify M_1 and M_2 by $\pm 1\%$) in order to simulate uncertainty in the model parameters. Which setup has the best tracking performance (i.e., feedback, feedforward, feedback + feedforward)?
2. (20 points) Estimate the resistance value starting from a series of repeated current and voltage measurements. Assume the true relationship

$$u_0(t) = R_0 i_0(t), \quad t = 1, 2, \dots, N,$$

where u_0, i_0 are the exact (noise-free) voltage and current.

Two different voltmeters are used, producing two data sets: one with low noise and one with high noise.

- **Data generation.** Generate an experiment with N measurements where i_0 is uniformly distributed in $[-0.01, 0.01]$ A and $R_0 = 1250 \, \Omega$. The current is measured without errors, while the voltage measured by the two voltmeters is disturbed by independent, zero-mean, normally distributed noise n_u :

$$n_u \sim \mathcal{N}(0, \sigma_u^2), \quad \sigma_u^2 = \begin{cases} 1, & \text{(good voltmeter),} \\ 16, & \text{(bad voltmeter).} \end{cases}$$

Thus, with the measured signals

$$i(t) = i_0(t), \quad u(t) = u_0(t) + n_u(t), \quad t = 1, 2, \dots, 2N,$$

the first N samples come from the good voltmeter and the next N from the bad one.

- (a) (5 points) **Weighted least-squares (WLS).** Derive the WLS estimate $\hat{R}$ of R by minimizing

$$V_{\text{WLS}} = \frac{1}{2N} \sum_{t=1}^{2N} \frac{(u(t) - R i(t))^2}{w(t)},$$

where $w(t)$ is the variance of the voltage noise corresponding to sample t . Also, derive the **unweighted** least squares estimator, which minimizes,

$$V_{LS} = \frac{1}{2N} \sum_{t=1}^{2N} (u(t) - Ri(t))^2 \quad (5)$$

Ans: To find the optimal $\hat{R}_{WLS}$, we differentiate with respect to R and set the expression equal to 0, to find the given R .

$$\begin{aligned} \frac{\partial V_{WLS}}{\partial R} &= 2 \cdot \frac{1}{2N} \sum_{t=1}^{2N} \frac{i(u - Ri)}{w} \\ &= \frac{1}{N} \sum_{t=1}^{2N} \frac{i u}{w} - R \frac{1}{N} \sum_{t=1}^{2N} \frac{i^2}{w} \\ \frac{1}{N} \sum_{t=1}^{2N} \frac{i u}{w} &= R \frac{1}{N} \sum_{t=1}^{2N} \frac{i^2}{w} \\ \hat{R}_{WLS} &= \frac{\sum_{t=1}^{2N} \frac{i(t)u(t)}{w(t)}}{\sum_{t=1}^{2N} \frac{i(t)^2}{w(t)}} \end{aligned}$$

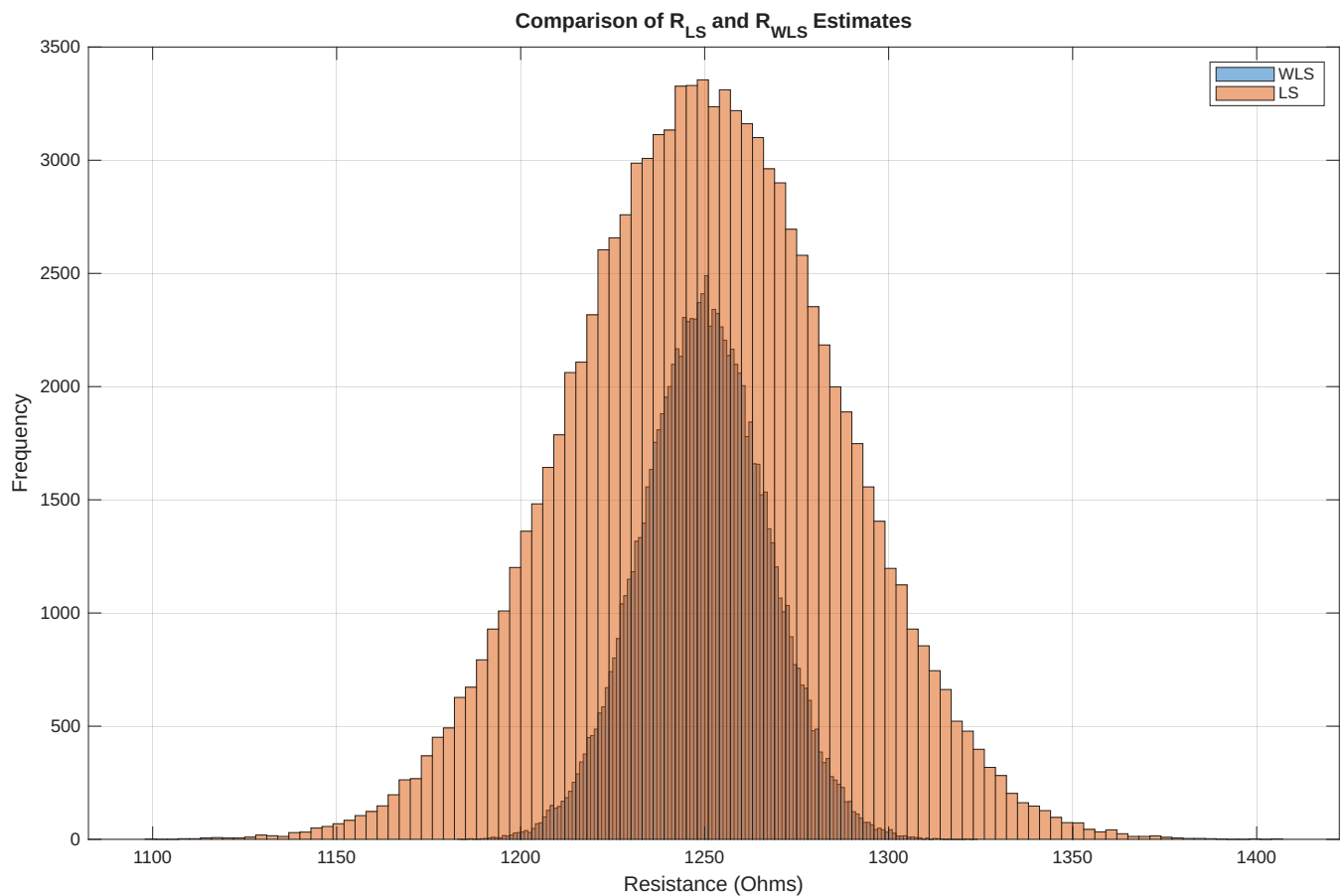
And for the LS method, without the weights added, we get

$$\begin{aligned} \frac{\partial V_{LS}}{\partial R} &= 2 \cdot \frac{1}{2N} \sum_{t=1}^{2N} i(u - Ri) \\ &= \frac{1}{N} \sum_{t=1}^{2N} i u - R \frac{1}{N} \sum_{t=1}^{2N} i^2 \\ \frac{1}{N} \sum_{t=1}^{2N} i u &= R \frac{1}{N} \sum_{t=1}^{2N} i^2 \\ \hat{R}_{LS} &= \frac{\sum_{t=1}^{2N} i u}{\sum_{t=1}^{2N} i^2} \end{aligned}$$

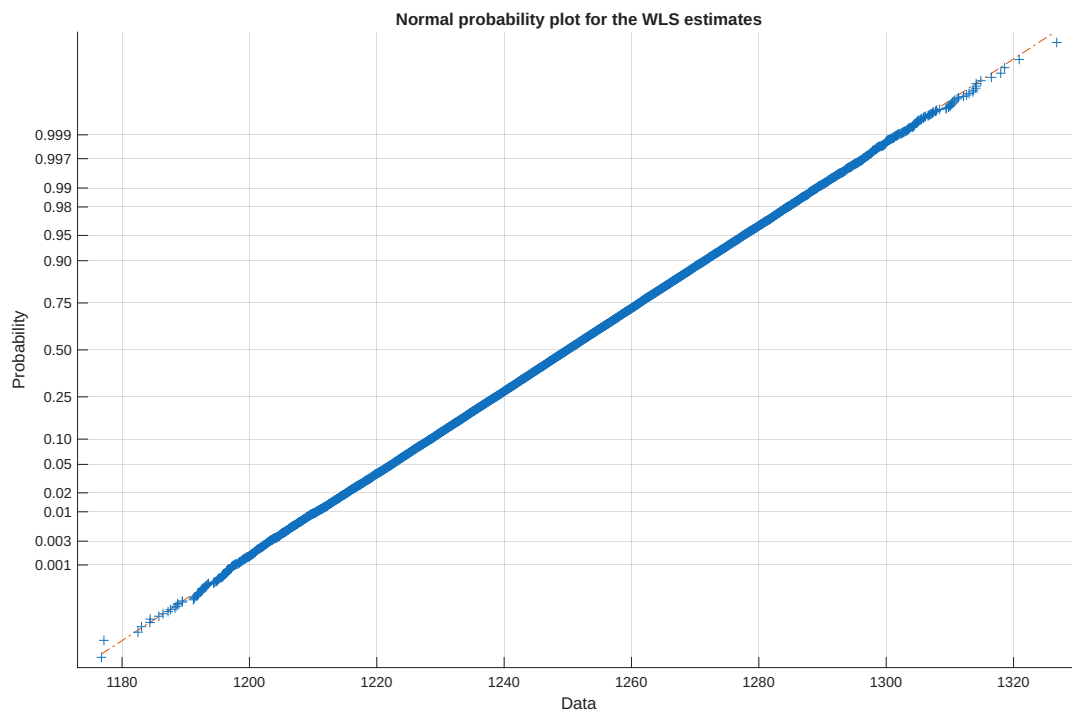
And the question should be answered

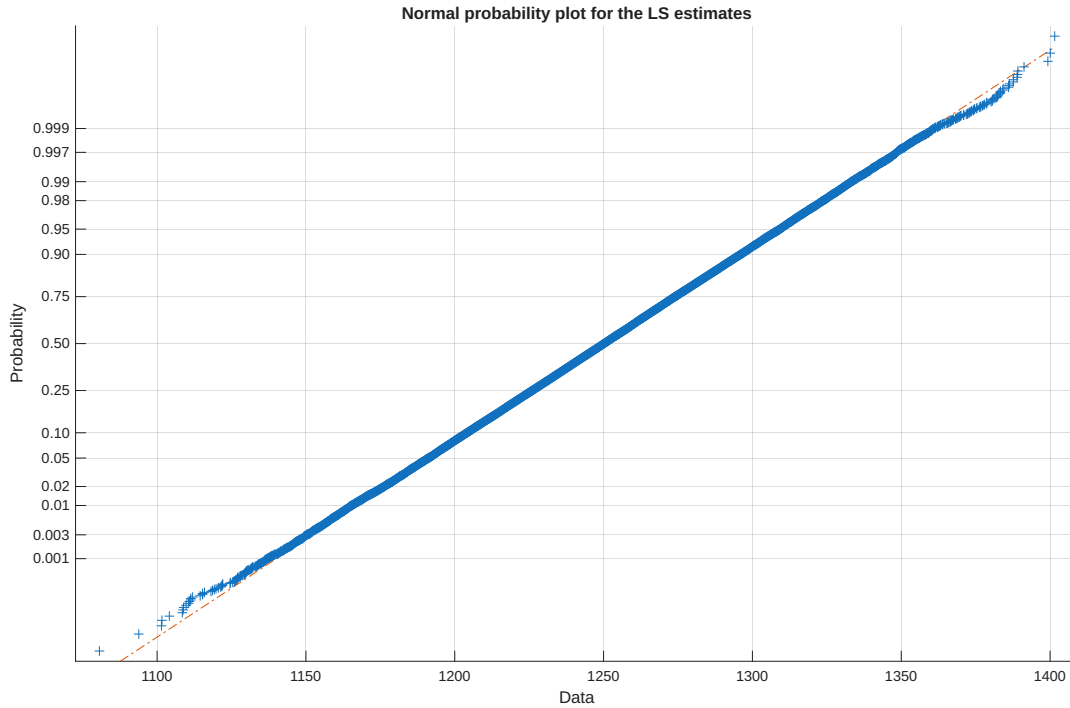
- (b) (5 points) Set $N = 100$. Compute an estimate $\hat{R}$ 10^5 times, using different data each time (according to the Data Generation procedure above), for both the LS and WLS procedures. Plot the resulting values of $\hat{R}$ in two histograms with an appropriate number of bins, one histogram for WLS and one for LS. Discuss the results.

Ans: There should be no surprise that the WLS estimates of the Ohms provide the most accurate ones. This happens because we have two voltmeters, and we give the voltmeter we trust the most, the most weight. This weight is given as the variance of the voltage noise ordered by the voltmeters.



The estimation of $\hat{R}$ over all the 10^5 experiments, leads to the figure above, which looks normally distributed, and indeed it obviously is, from the errors from the 2 voltmeters that are normally distributed





Basically, if the two arrays for the estimates, are closer to the straight line with slope $\frac{1}{1320-1180} = \frac{1}{140}$, the more normal it is distributed. The plots above confirms our belief.

- (c) (10 points) Derive expressions for the variance of the WLS and the LS estimators. Compute the values of each based on the model parameters listed in the Data Generation procedure. Then, compute the sample variance of the estimates $\hat{R}$ in your experiments in part (a). Compare these to the expressions that you derived/computed. What would happen as the number of experiments increases from 10^5 ?

Ans: To derive the variance of the estimators, we take the $\hat{R}_{LS}$ and $\hat{R}_{WLS}$ and set them as inputs to the function $V(x)$

$$\begin{aligned}
 V(\hat{R}_{WLS}) &= V\left(\frac{\sum_{t=1}^{2N} \frac{i(t)u(t)}{w(t)}}{\sum_{t=1}^{2N} \frac{i(t)^2}{w(t)}}\right) \\
 &= V\left(\frac{\sum_{t=1}^{2N} \frac{i(t)(R_0 i_0(t) + n_u(t))}{w(t)}}{\sum_{t=1}^{2N} \frac{i(t)^2}{w(t)}}\right) \\
 &= V\left(\frac{\sum_{t=1}^{2N} \frac{i(t)R_0 i_0(t)}{w(t)}}{\sum_{t=1}^{2N} \frac{i(t)^2}{w(t)}} + \frac{\sum_{t=1}^{2N} \frac{i(t)n_u(t)}{w(t)}}{\sum_{t=1}^{2N} \frac{i(t)^2}{w(t)}}\right)
 \end{aligned}$$

Since $i(t) = i_0(t)$ is the **true** current, we can substitute that into the first fraction and get

$$V(\hat{R}_{WLS}) = V(R_0) + \frac{\sum_{t=1}^{2N} \frac{i(t)^2 V(n_u(t))}{w(t)^2}}{\left(\sum_{t=1}^{2N} \frac{i(t)^2}{w(t)}\right)^2}$$

And since $V(n_u(t)) = w(t)$, we get

$$\begin{aligned} V(\hat{R}_{WLS}) &= \frac{\sum_{t=1}^{2N} \frac{i(t)^2}{w(t)}}{\sum_{t=1}^{2N} \frac{i(t)^4}{w(t)^2}} \\ &= \frac{1}{\sum_{t=1}^{2N} \frac{i(t)^2}{w(t)}} \end{aligned}$$

Which is the final expression. Its important to note that for a linear equation $a+bx$, then $V(a+bx) = b^2V(x)$ where x is a random variable, which is why the R_0 term disappears

Now for the derivation of $\hat{R}_{LS}$

$$\begin{aligned} V(\hat{R}_{LS}) &= V\left(\frac{\sum_{t=1}^{2N} i(t)u(t)}{\sum_{t=1}^{2N} i(t)^2}\right) \\ &= V\left(\sum_{t=1}^{2N} \frac{i(t)(R_0 i_0(t) + n_u(t))}{i(t)^2}\right) \\ &= V\left(\sum_{t=1}^{2N} \frac{i(t)R_0 + i(t)^2 n_u(t)}{i(t)^2}\right) \\ &= V(R_0) + \sum_{t=1}^{2N} \frac{i(t)^2 V(n_u(t))}{i(t)^4} \\ &= \frac{1}{\sum_{t=1}^{2N} \frac{i(t)^2}{w(t)}} \end{aligned}$$

from the same reasoning that $V(n_u(t)) = w(t)$

3. (20 points) **Least-Squares Identification of a One-Parameter Linear System.** Consider the discrete-time system

$$y_k = a y_{k-1} + b u_{k-1} + e_k, \quad k = 2, \dots, N,$$

where

- y_k is the measured output,
- u_k is the known input,
- e_k is a zero-mean white noise sequence,
- a, b are unknown parameters to be estimated.

Stack all samples $k = 2, \dots, N$ to obtain

$$\mathbf{y} = \Phi \theta + \mathbf{e},$$

with

$$\Phi = \begin{bmatrix} y_1 & u_1 \\ y_2 & u_2 \\ \vdots & \vdots \\ y_{N-1} & u_{N-1} \end{bmatrix}, \quad \theta = \begin{bmatrix} a \\ b \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_2 \\ y_3 \\ \vdots \\ y_N \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} e_2 \\ e_3 \\ \vdots \\ e_N \end{bmatrix}. \quad (6)$$

(a) (5 points) By minimizing the sum of squared residuals

$$V(a, b) = \sum_{k=2}^N (y_k - a y_{k-1} - b u_{k-1})^2,$$

show that the optimal parameters satisfy the *normal equations*

$$\Phi^T \Phi \hat{\theta} = \Phi^T \mathbf{y},$$

leading to the closed-form least-squares solution

$$\boxed{\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}.} \quad (7)$$

Ans: To derive the optimal parameters that satisfy the normal equations, we see first that from the exercise, we are given that we have dropped the error terms in the sum. And when we stack all the y_k, u_k , and all a, b that shall be estimated, we get the sum that's given in the exercise. In quadratic form, this turns to

$$\begin{aligned} r &= y - \Phi\theta \\ V(\theta) &= r^T r \\ &= (y - \Phi\theta)^T (y - \Phi\theta) \\ &= y^T y - y^T \Phi\theta - (\Phi\theta)^T y + (\Phi\theta)^T (\Phi\theta) \\ &= y^T y - 2\theta^T \Phi^T y + \theta^T \Phi^T \Phi \theta \end{aligned}$$

And now we have to find the optimal parameters a, b that find the minimized sum of squared residuals

$$\begin{aligned} \frac{\partial V(\theta)}{\partial \theta} &= 0 - 2\Phi^T y + 2\Phi^T \Phi \theta = 0 \\ 2(\Phi^T \Phi) \hat{\theta} &= 2\Phi^T y \\ \hat{\theta} &= (\Phi^T \Phi)^{-1} \Phi^T y \end{aligned}$$

This assumes of course that $\Phi^T \Phi$ is invertible. I will get to this more in 3 c). But all in all, I think the exercise is solved.

(b) (10 points) Set $N = 100$, $a = 0.2$ and $b = -0.25$, and $e_k \sim \mathcal{N}(0, 0.5)$, and consider the following input sequences:

- a random input that is 1 half the time, and -1 half the time:

$$u_k \sim \begin{cases} 1 & p = 0.5 \\ -1 & p = -0.5 \end{cases} \quad (8)$$

- a random uniform input:

$$u_k \sim \text{randu}[-1,1] \quad (9)$$

- a gaussian input with mean zero and variance 1:

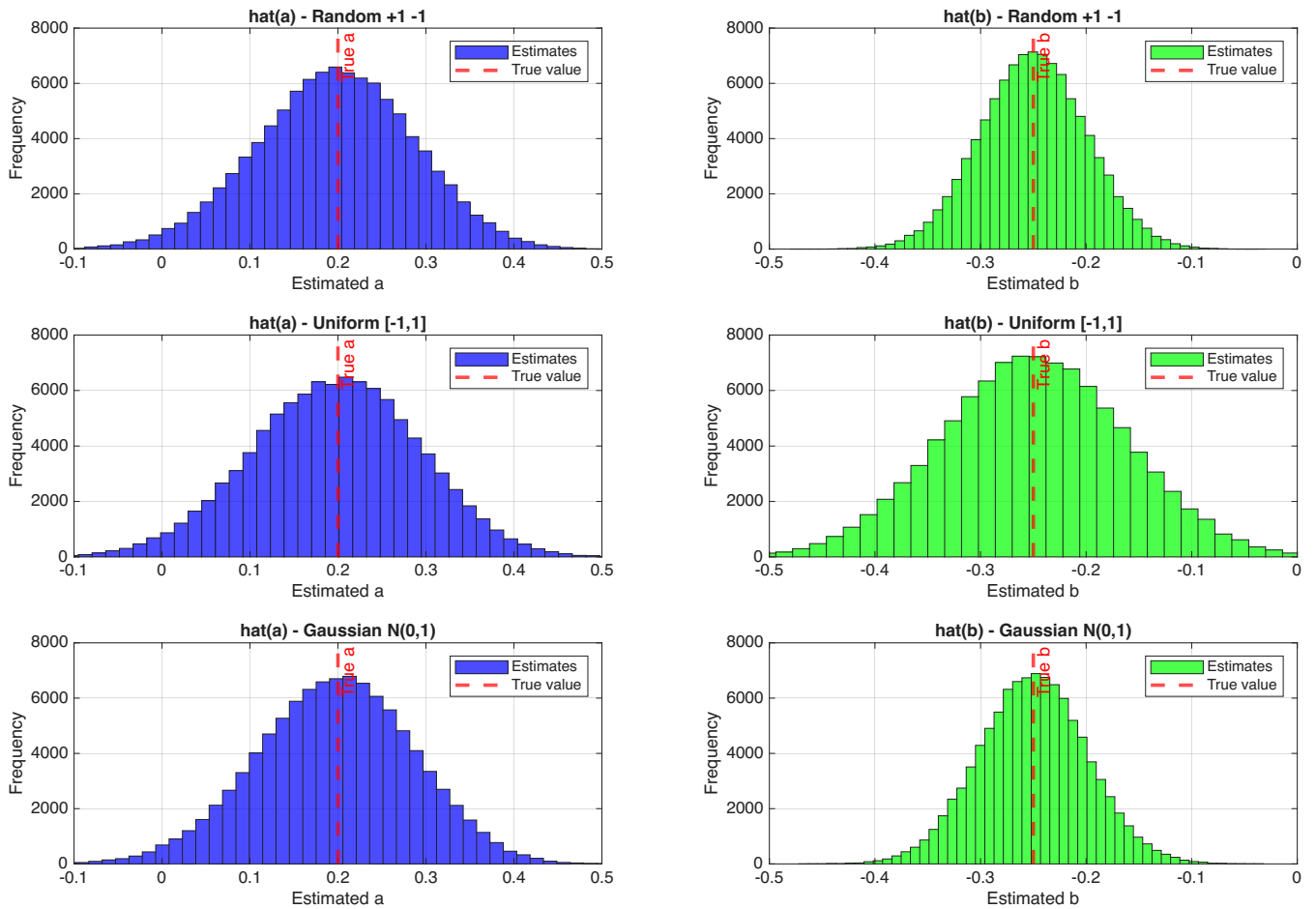
$$u_k \sim \mathcal{N}(0, 1) \quad (10)$$

For each input type, perform the following:

- Generate $\mathbf{y}$ using Equation (6)
- Generate $\hat{\theta}$ using Equation (7)
- Repeat this process 10^5 times

Plot histograms for $\hat{a}$ and for $\hat{b}$ for the different input types on the same plot.
Which input type yields the lowest variance estimate?

Ans: Here is the histogram plots for the different a and b:



I will refer to the different control series generation processes like the following:

- **Method 1**: a random input that is 1 half the time, and -1 half the time:

$$u_k \sim \begin{cases} 1 & p = 0.5 \\ -1 & p = -0.5 \end{cases} \quad (11)$$

- **Method 2**: a random uniform input:

$$u_k \sim \text{randu}[-1,1] \quad (12)$$

- **Method 3**: a gaussian input with mean zero and variance 1:

$$u_k \sim \mathcal{N}(0, 1) \quad (13)$$

I also gather that the empirical variance of the $\hat{a}$ and $\hat{b}$ arrays vary between the method 1 and method 3. The following table displays the variance of said arrays, where column i is either the $\hat{a}$ array or the $\hat{b}$ array, and the column i is $method_i$

Input Type	Var($\hat{a}$)	Var($\hat{b}$)
Method 1	0.007833	0.002547
Method 2	0.009062	0.007751
Method 3	0.007848	0.002587

From this, we see that method 2 is without a doubt the worst. It is relatively close between method 1 and method 2 having the lowest variance implication on the coefficient estimates $\hat{a}$ and $\hat{b}$.

It comes to the point where it differs between each time I run the code.

(c) (5 points) Discussion

1. Under what conditions is $\Phi^T \Phi$ invertible?

Ans: $\Phi^T \Phi$ is invertible if and only if the following is satisfied; The columns of $\Phi^T \Phi$ have full rank. The only case where this would not happen, was if the controls $u_k = 0 \forall k$. This is unlikely in practice

2. What happens if the input u_k is constant?

Ans: If one of the controls are constant, I don't see any much affection on the series y. But if $u_k = 0 \forall k$ then the matrix $\Phi^T \Phi$ is non-invertible. I mentioned this earlier. In this case, the estimates $\hat{a}$ and $\hat{b}$ are all zero. This is because the equation

$$\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T y$$

would not exist, due non-invertible property of the $\Phi^T \Phi$ matrix

3. How would correlated noise e_k affect the bias of the estimates?

Ans: FDGDS